

# Ayoub ABIDI

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## Summary

An energetic, creative, and open-minded ICT engineering student at the National Engineering School of Tunis, passionate about data science and artificial intelligence. Driven by curiosity and problem-solving, with a strong desire to turn data into meaningful insights and innovative solutions through analytical and logical thinking.

## Education

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|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>National Engineering School of Tunis (ENIT)</b><br><i>Master's Degree in Information Processing Technology</i>                               | <i>Sep 2025 - Present</i>  |
| <b>National Engineering School of Tunis (ENIT)</b><br><i>National Engineering Degree in Telecommunications - Specialization in Data Science</i> | <i>Sep 2023 - Present</i>  |
| <b>Preparatory Institute for Engineering Studies of Tunis (IPEIT)</b><br><i>Pre-Engineering Studies - Physics and Chemistry Section</i>         | <i>Sep 2021 - Jun 2023</i> |

## Experience

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| <b>Software &amp; AI Engineering Intern</b><br><i>Talan Consulting</i>                                                                                                                                                                                                                                      | <i>Jun 2025 – Aug 2025</i> |
| <ul style="list-style-type: none"><li>Developed full-stack intelligent log analysis platform with automated parsing, anomaly detection, and correction recommendation.</li><li>Implemented an AI layer using Generative AI and RAG techniques, with a Spring Boot backend and a Next.js frontend.</li></ul> |                            |
| <b>Research Intern (PFA2) - Deep Learning for EEG Classification</b><br><i>RISC Laboratory, ENIT</i>                                                                                                                                                                                                        | <i>Oct 2024 – Apr 2025</i> |
| <ul style="list-style-type: none"><li>Conducted comparative study of deep learning models (ANN, CNN, CNN-LSTM) for epileptic patient classification from EEG signals.</li><li>Developed Streamlit interface for real-time EEG classification supporting clinical decision-making.</li></ul>                 |                            |

## Projects

**Multimodal RAG for Tunisian Sign Language Translation** (*Multimodal RAG, VLM, LLM*) : Built a real-time AI system that translates Tunisian Sign Language into text and speech using CLIP-based visual embeddings, a custom vector database, and LLM-powered Retrieval-Augmented Generation.

**Epileptic Seizure Prediction** (*Deep Learning / Signal Processing*) : Developed a deep learning pipeline to predict epileptic seizures from multi-channel EEG signals using the TUH EEG Seizure Corpus.

**Text-to-Speech Synthesis** (*Deep Learning, Speech Synthesis, Neural Vocoders*) : Developed a TTS system generating natural and expressive speech from text, using pre-trained models and neural vocoders, with optional multi-speaker support.

**Credit Line Adjuster using Reinforcement Learning** (*RL / Finance*) : Q-Learning agent for dynamic credit line adjustment with interactive dashboard for performance visualization.

## Technical Skills

**Programming Languages:** Python, C, C++, C#, Java, Arduino, PL/SQL    **Web Development:** HTML/CSS, JavaScript, TypeScript

**Frameworks:** .NET, Spring Boot, Next.js, Streamlit, Chainlit    **Data Analysis & ML:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV

**Generative AI & LLMs:** Hugging Face, Groq API, Ollama, LangChain, CLIP (VLM), Mistral, LLaMA, Gemma, Chainlit

**Databases:** MySQL, PostgreSQL, SQLite3    **Others:** Docker, CI/CD, Git, GitHub, Linux Administration

## Languages

|                |        |                 |    |
|----------------|--------|-----------------|----|
| <b>Arabic:</b> | Native | <b>English:</b> | B2 |
| <b>French:</b> | B2     | <b>German:</b>  | A2 |

## Associative Life

**Vice President**, IEEE ENIT Student Branch (2025) - **Treasurer** IEEE ENIT Student Branch (2024) -  
**Vice Chair**, IEEE WIE Annual Congress of Tunisia 4.0 - **Project Manager**, Tunisia Engineering Day 7.0 (2024)